Stable LatentMoE

Deep Dive into Kimi K3 — Chapter 8

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Foundation Books Project

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Goal: type the latent forward, bound its activations, derive Quantile Balancing from a balanced assignment, estimate it with pooled histograms, and keep each measurement in scope.

Roadmap

A typed latent forward

Balancing the whole step

Ledgers and evidence

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- Membership by TopK($s_t + b^{(n)}$); weights from selected raw s_t.

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- SiTU bounds each pre-W_o coordinate by $\beta_1\beta_2 = 100$; RMSNorm acts once, on the weighted aggregate.

Balancing the whole step

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- Histogram estimator: all-reduce integer counts, then recover each rank within one bin width — never average local quantiles.

Ledgers and evidence

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Diagnose in dependency order. Next: Chapter 9's MoonViT and Native Multimodality.